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Abstract

We study the ensemble performance of biometric authentication systems, based on secret key generation, which work as follows. In the enrollment stage, an individual provides a biometric signal that is mapped into a secret key and a helper message, the former being prepared to become available to the system at a later time (for authentication), and the latter is stored in a public database. When an authorized user requests authentication, claiming his/her identity as one of the subscribers, s/he has to provide a biometric signal again, and then the system, which retrieves also the helper message of the claimed subscriber, produces an estimate of the secret key, that is finally compared to the secret key of the claimed user. In case of a match, the authentication request is approved, otherwise, it is rejected. Referring to an ensemble of systems based on Slepian–Wolf binning, we provide a detailed analysis of the false–reject and false–accept probabilities, for a wide class of stochastic decoders. We also comment on the security for the typical code in the ensemble.

Index Terms: biometric security, Slepian-Wolf coding, random binning, error exponents, secrete key generation.

I. Introduction

We consider a biometric authentication system that is described in [7, Sections 2.2–2.6], which is based on the notion of secret key generation and sharing due to Maurer [8] and Ahlswede and Csiszár [1], [2]. Specifically, such a system works as follows. In the enrollment stage, an individual which subscribes to the system provides a biometric signal, $\boldsymbol{X} = (X_1, X_2, \ldots, X_n)$. The system receives this signal and generates (using its encoder) two outputs in response. The first output is a secret key, $\boldsymbol{S}$, at rate $R_{\rm s}$ and the second is a helper message, $\boldsymbol{W}$, at rate $R_{\rm w}$. The secret

key is prepared in order to be used by the system later, at the authentication stage. The helper message is stored in a public database. When an authorized user (a subscriber) wishes to sign in, claiming his/her identity as one of the existing subscribers, s/he is requested to provide again his/her biometric signal, $\boldsymbol{Y} = (Y_1, \ldots, Y_n)$ (correlated to $\boldsymbol{X}$, if indeed from the same individual, or independent, if not). The system then retrieves the helper message $\boldsymbol{W}$ of the claimed subscriber, and responds (using its decoder) by estimating the secret key, $\hat{\boldsymbol{S}}$ (based on $(\boldsymbol{Y}, \boldsymbol{W})$), and comparing it to the secret key of the claimed user, $\boldsymbol{S}$. In case of a match, access to the system is granted, otherwise, it is denied.

In [7, Sect. 2.3], achievable rate pairs $(R_{\rm s}, R_{\rm w})$ were found for the existence of systems (encoders and decoders) that satisfy the following three requirements in the large n limit: (i) arbitrarily small false–reject (FR) probability, (ii) arbitrarily small false–accept (FA) probability, and (iii) arbitrarily small leakage between the secret message and the helper message, in terms of the asymptotic normalized mutual information, $I(\boldsymbol{S};\boldsymbol{W})/n$. In particular, Theorem 2.1 of [7] asserts that when $(\boldsymbol{X}, \boldsymbol{Y})$ are drawn from a discrete memoryless source (DMS), generating independent copies of a correlated pair $(X, Y) \sim P_{XY}$, the maximum achievable key rate, $R_{\rm s}$, under the above constraints, is given by the single–letter mutual information, $I(X;Y)$. It then follows that $R_{\rm w}$ must lie in the range $H(X|Y) < R_{\rm w} < H(X) - R_{\rm s}$, where the conditional entropy in the lower limit is essential for reliable identification of an authorized subscriber (small FR probability) and the upper limit is essential for the security requirement. These limitations already guarantee that $R_{\rm w} < H(X)$, which is essential for keeping the FA probability vanishingly small for large n.

As in many proofs of direct coding theorems in the information theory literature, in the achievability part of [7, Theorem 2.1] too, the analyses of the error probabilities (in this case, the FA and the FR probabilities) are very rough – they are merely good enough to prove the achievability of the desired coding rates in the simplest possible manner. However, these are poor estimates of the achievable FR and FA probabilities themselves when these are considered to be the relevant performance metrics for given $R_{\rm s}$ and $R_{\rm w}$.

The purpose of this paper is to provide sharper evaluations of the ensemble performance of the FA and the FR probabilities. In particular, referring to an ensemble of systems based on Slepian–Wolf binning, we provide detailed analyses of the exponential behavior of the FR probability, for

a wide class of stochastic decoders, which includes the respective maximum a posteriori (MAP) decoder as a special case. An expurgated bound is provided as well and discussed quite in detail. For the FA probability, we analyze the ensemble performance of the MAP decoder and provide some intuition concerning its behavior. We also comment on the security of the code for the typical code in the ensemble.

The paper is organized as follows. In Section II, we establish the notation conventions. In Section III, we formalize the setup and spell out the objectives. In Section IV, we present and discuss the random coding FR exponent and an expurgated bound. In Section V, we derive the random coding FA exponent, and finally, in Section VI, we discuss the leakage of the typical code.

II. Notation Conventions

Throughout the paper, random variables will be denoted by capital letters, specific values they may take will be denoted by the corresponding lower case letters, and their alphabets will be denoted by calligraphic letters. Random vectors and their realizations will be denoted, respectively, by capital letters and the corresponding lower case letters, both in the bold face font. Their alphabets will be superscripted by their dimensions. For example, the random vector $\boldsymbol{X} = (X_1, \ldots, X_n)$, ($n$ – positive integer) may take a specific vector value $\boldsymbol{x} = (x_1, \ldots, x_n)$ in $\mathcal{X}^n$, the n–th order Cartesian power of $\mathcal{X}$, which is the alphabet of each component of this vector. Sources and channels will be denoted by the letter P or Q, subscripted by the names of the relevant random variables/vectors and their conditionings, if applicable, following the standard notation conventions, e.g., Q_X, $P_{Y|X}$, and so on. When there is no room for ambiguity, these subscripts will be omitted. The probability of an event $\mathcal{G}$ will be denoted by $\Pr\{\mathcal{G}\}$, and the expectation operator with respect to (w.r.t.) a probability distribution P will be denoted by $\boldsymbol{E}_P\{\cdot\}$. Again, the subscript will be omitted if the underlying probability distribution is clear from the context. The entropy of a generic distribution Q on $\mathcal{X}$ will be denoted by $H_Q(X)$. For two positive sequences a_n and b_n, the notation $a_n \doteq b_n$ will stand for equality in the exponential scale, that is, $\lim_{n\to\infty} \frac{1}{n} \log \frac{a_n}{b_n} = 0$. Similarly, $a_n \dot{\leq} b_n$ means that $\limsup_{n\to\infty} \frac{1}{n} \log \frac{a_n}{b_n} \leq 0$, and so on. The indicator function of an event $\mathcal{G}$ will be denoted by $\mathcal{I}\{\mathcal{G}\}$. The notation $[x]_+$ will stand for $\max\{0, x\}$.

The empirical distribution of a sequence $\boldsymbol{x} \in \mathcal{X}^n$, which will be denoted by $\hat{P}_{\boldsymbol{x}}$, is the vector of

relative frequencies $\hat{P}_{\boldsymbol{x}}(x)$ of each symbol $x \in \mathcal{X}$ in $\boldsymbol{x}$. The type class of $\boldsymbol{x} \in \mathcal{X}^n$, denoted $\mathcal{T}(\hat{P}_{\boldsymbol{x}})$, is the set of all vectors $\boldsymbol{x}'$ with $\hat{P}_{\boldsymbol{x}'} = \hat{P}_{\boldsymbol{x}}$. Information measures associated with empirical distributions will be denoted with 'hats' and will be subscripted by the sequences from which they are induced. For example, the entropy associated with $\hat{P}_{\boldsymbol{x}}$, which is the empirical entropy of $\boldsymbol{x}$, will be denoted by $\hat{H}_{\boldsymbol{x}}(X)$. Similar conventions will apply to the joint empirical distribution, the joint type class, the conditional empirical distributions and the conditional type classes associated with pairs (and multiples) of sequences of length n. Accordingly, $\hat{P}_{\boldsymbol{xy}}$ will be the joint empirical distribution of $(\boldsymbol{x}, \boldsymbol{y}) = \{(x_i, y_i)\}_{i=1}^n$, and $\mathcal{T}(\hat{P}_{\boldsymbol{xy}})$ will denote the joint type class of $(\boldsymbol{x}, \boldsymbol{y})$. Similarly, $\mathcal{T}(\hat{P}_{\boldsymbol{x}|\boldsymbol{y}}|\boldsymbol{y})$ will stand for the conditional type class of $\boldsymbol{x}$ given $\boldsymbol{y}$, $\hat{H}_{\boldsymbol{xy}}(X, Y)$ will designate the empirical joint entropy of $\boldsymbol{x}$ and $\boldsymbol{y}$, $\hat{H}_{\boldsymbol{xy}}(X|Y)$ will be the empirical conditional entropy, $\hat{I}_{\boldsymbol{xy}}(X;Y)$ will denote empirical mutual information, and so on. We will also use similar rules of notation in the context of a generic distribution, Q_{XY} (or Q, for short): we use $\mathcal{T}(Q_X)$ for the type class of sequences with empirical distribution Q_X, $H_Q(X)$ – for the corresponding empirical entropy, $\mathcal{T}(Q_{XY})$ – for the joint type class x, $T(Q_{X|Y}|\boldsymbol{y})$ – for the conditional type class of $\boldsymbol{x}$ given $\boldsymbol{y}$, $H_Q(X, Y)$ – for the joint empirical entropy, $H_Q(X|Y)$ – for the conditional empirical entropy, $I_Q(X;Y)$ – for the empirical mutual information, and so on. We will also use the customary notation for the weighted divergence,

$$D(Q_{Y|X}\|P_{Y|X}|Q_X) = \sum_{x\in\mathcal{X}} Q_X(x) \sum_{y\in\mathcal{Y}} Q_{Y|X}(y|x) \log \frac{Q_{Y|X}(y|x)}{P_{Y|X}(y|x)}. \tag{1}$$

III. Setup and Objectives

Consider the following system model for biometric identification. An *enrollment source* sequence, $\boldsymbol{x} = (x_1, \ldots, x_n)$, which is a realization of the random vector $\boldsymbol{X} = (X_1, \ldots, X_n)$, that emerges from a discrete memoryless source (DMS), P_X, with a finite alphabet $\mathcal{X}$, is fed into an *enrollment encoder*, $\mathcal{E}$, that produces two outputs: a secret key, $\boldsymbol{s}$ (a realization of a random variable $\boldsymbol{S}$), and a helper message, $\boldsymbol{w}$ (a realization of $\boldsymbol{W}$), taking on values in finite alphabets, $\mathcal{S}_n = \{0, 1, \ldots, e^{nR_{\rm s}}\}$ and $\mathcal{W}_n = \{0, 1, \ldots, e^{nR_{\rm w}}\}$, respectively, where $R_{\rm s}$ is the *secret–key rate*, and $R_{\rm w}$ is the *helper–message rate*. This encoding operation designates the enrollment stage.

We consider the ensemble of enrollment encoders, $\{\mathcal{E}\}$, generated by *random binning*, where for each source vector $\boldsymbol{x} \in \mathcal{X}$, one selects independently at random, both a secret key and a helper

message, under the uniform distributions across $\mathcal{S}_n$ and $\mathcal{W}_n$, respectively. In other words, denoting by $\boldsymbol{w} = f(\boldsymbol{x})$ and $\boldsymbol{w} = f(\boldsymbol{x})$, the randomly selected bin assignments for both outputs, it is assumed that the $2|\mathcal{X}|^n$ random variables $\{f(\boldsymbol{x}), g(\boldsymbol{x})\}_{\boldsymbol{x}\in\mathcal{X}^n}$ are all mutually independent.

The *authentication decoder*, $\mathcal{A}$, which is aware of the randomly selected encoder, $\mathcal{E}$, is fed by two inputs: the helper message $\boldsymbol{w}$ and an *authentication source* sequence, $\boldsymbol{y} = (y_1, \ldots, y_n)$ (a realization of $\boldsymbol{Y} = (Y_1, \ldots, Y_n)$), that is produced at the output of a discrete memoryless channel (DMC), $P_{Y|X}$, with a finite output alphabet $\mathcal{Y}$, that is fed by $\boldsymbol{x}$. The output of the authentication decoder is $\hat{\boldsymbol{s}} = U(\boldsymbol{y}, \boldsymbol{w})$ (a realization of $\hat{\boldsymbol{S}}$), which is an estimate (possibly, randomized) of the secret key, $\boldsymbol{s}$. If $\hat{\boldsymbol{s}} = \boldsymbol{s}$, access to the system is granted, otherwise, it is denied. This decoding operation stands for the authentication stage.

The optimal estimator of $\boldsymbol{s}$, based on $(\boldsymbol{y}, \boldsymbol{w})$, in the sense of minimum FR probability, $\Pr\{\hat{S} \neq \boldsymbol{S}\}$, is the maximum a posteriori probability (MAP) estimator, given by

$$\hat{\boldsymbol{s}}_{\mathrm{MAP}} = U(\boldsymbol{y}, \boldsymbol{w}) \stackrel{\Delta}{=} \arg\max_{\boldsymbol{s}} P(\boldsymbol{s}, \boldsymbol{w}|\boldsymbol{y}) = \arg\max_{\boldsymbol{s}} \sum_{\boldsymbol{x}\in\mathcal{X}^n} P(\boldsymbol{x}|\boldsymbol{y}) \cdot \mathcal{I}\{f(\boldsymbol{x}) = \boldsymbol{w}\} \cdot \mathcal{I}\{g(\boldsymbol{x}) = \boldsymbol{s}\}, \quad (2)$$

where $P(\boldsymbol{x}|\boldsymbol{y})$ (shorthand notation for $P_{\boldsymbol{X}|\boldsymbol{Y}}(\boldsymbol{x}|\boldsymbol{y})$) is the posterior probability of $\boldsymbol{X} = \boldsymbol{x}$ given $\boldsymbol{Y} = \boldsymbol{y}$, that is induced by the product distribution, P_{XY} (and the subscript XY will sometimes be suppressed for simplicity, when there is no risk of compromising clarity).

In this paper, we expand the scope and study a more general class of decoders. This is a class of generalized stochastic likelihood decoders [10], [13], [14], [16], where the decoder randomly selects its estimate $\hat{\boldsymbol{s}}$ according to the posterior distribution

$$\tilde{P}(\boldsymbol{s}|\boldsymbol{y}, \boldsymbol{w}) = \frac{\sum_{\boldsymbol{x}\in\mathcal{X}^n} \exp\{na(\hat{P}_{\boldsymbol{x}\boldsymbol{y}})\} \cdot \mathcal{I}\{f(\boldsymbol{x}) = \boldsymbol{w}\} \cdot \mathcal{I}\{g(\boldsymbol{x}) = \boldsymbol{s}\}}{\sum_{\boldsymbol{x}\in\mathcal{X}^n} \exp\{na(\hat{P}_{\boldsymbol{x}\boldsymbol{y}})\} \cdot \mathcal{I}\{f(\boldsymbol{x}) = \boldsymbol{w}\}}, \quad (3)$$

where the function $a(\cdot)$, henceforth referred to as the *decoding metric*, is an arbitrary continuous function of the joint empirical distribution $\hat{P}_{\boldsymbol{x}\boldsymbol{y}}$. Throughout the sequel, we will refer to the numerator of the r.h.s. as $\tilde{P}(\boldsymbol{s}, \boldsymbol{w}|\boldsymbol{y})$, and to the denominator as $\tilde{P}(\boldsymbol{w}|\boldsymbol{y})$. For

$$a(\hat{P}_{\boldsymbol{x}\boldsymbol{y}}) = \sum_{x\in\mathcal{X}} \sum_{y\in\mathcal{Y}} \hat{P}_{\boldsymbol{x}\boldsymbol{y}}(x, y) \ln P(x|y), \quad (4)$$

we have the ordinary likelihood decoder in spirit of [13], [14], [16]. For

$$a(\hat{P}_{\boldsymbol{x}\boldsymbol{y}}) = \beta \sum_{x\in\mathcal{X}} \sum_{y\in\mathcal{Y}} \hat{P}_{\boldsymbol{x}\boldsymbol{y}}(x, y) \ln P(x|y), \quad (5)$$

β being a free parameter (sometimes referred to as the inverse temperature parameter [12] due to the analogy in statistical mechanics), we extend this likelihood decoder to a parametric family of decoders, where β controls the skewedness of the posterior. In particular, $\beta \to \infty$ leads to the ordinary MAP decoder, $\hat{\boldsymbol{s}}_{\mathrm{MAP}}$. Other interesting choices are associated with mismatched metrics,

$$a(\hat{P}_{\boldsymbol{xy}}) = \sum_{x \in \mathcal{X}} \sum_{y \in \mathcal{Y}} \hat{P}_{\boldsymbol{xy}}(x, y) \ln P'(x|y), \tag{6}$$

P' being different from P, and

$$a(\hat{P}_{\boldsymbol{xy}}) = -\beta \hat{H}_{\boldsymbol{xy}}(X|Y), \tag{7}$$

which for $\beta \to \infty$ approaches the universal minimum entropy decoder (see also discussion around eqs. (5)–(7) of [10]).

An unauthorized user (i.e., an imposter), who claims for a given subscriber identity and wishes to break into the system, does not have the correlated biometric data $\boldsymbol{y}$. The best s/he can do is to estimate $\boldsymbol{s}$ based on the only data s/he has, which is the helper message $\boldsymbol{w}$, and then forges any fake biometric data $\tilde{\boldsymbol{y}}$, which together with $\boldsymbol{w}$, would cause the decoder to output this estimate of $\boldsymbol{s}$. More precisely, the imposter first estimates $\boldsymbol{s}$ according to

$$\tilde{\boldsymbol{s}} = V(\boldsymbol{w}) \stackrel{\Delta}{=} \arg\max_{\boldsymbol{s}} P(\boldsymbol{s}|\boldsymbol{w}) = \arg\max_{\boldsymbol{s}} \sum_{\boldsymbol{x} \in \mathcal{X}^n} P(\boldsymbol{x}) \cdot \mathcal{I}\{f(\boldsymbol{x}) = \boldsymbol{w}\} \cdot \mathcal{I}\{g(\boldsymbol{x}) = \boldsymbol{s}\}, \tag{8}$$

and then generates any $\tilde{\boldsymbol{y}} \in \mathcal{Y}^n$ such that $U(\tilde{\boldsymbol{y}}, \boldsymbol{w}) = \tilde{\boldsymbol{s}}$, and uses it as the biometric signal for authentication.

The objectives of the paper are to obtain: (i) ensemble–tight, exponential error bounds for the average FR probability, $\bar{P}_{\mathrm{FR}} = \Pr\{\hat{\boldsymbol{S}} \neq \boldsymbol{S}\}$, associated with the generalized stochastic likelihood decoder (3), as well as an expurgated bound following the methodology of [10, Theorem 2] (see also the correction [11]), and (ii) an exponential error bound for the average FA probability of (8), $\bar{P}_{\mathrm{FA}} = \Pr\{\tilde{\boldsymbol{S}} = \boldsymbol{S}\}$. Finally, we provide an outline of a derivation of the leakage, $I(\boldsymbol{S}; \boldsymbol{W})$, for a typical code, $\mathcal{E}$, in the large n limit.

IV. False–Reject Error Analysis

A. Random Coding Exponent

Consider the system configuration described in Section III, along with the generalized stochastic likelihood decoder (3). Define the functions

$$E(R_{\rm w}, Q_{X_0Y}) \stackrel{\Delta}{=} \min_{Q_{X|Y}} [R_{\rm w} - H_Q(X|Y) + [a(Q_{X_0Y}) - a(Q_{XY})]_+]_+ \tag{9}$$

and

$$E_{\rm r}^{\rm FR}(R_{\rm w}) \stackrel{\Delta}{=} \min_{Q_{X_0Y}} \{D(Q_{X_0Y}\|P_{XY}) + E(R_{\rm w}, Q_{X_0Y})\}. \tag{10}$$

Our first result is the following.

Theorem 1 *Consider the system configuration described in Section III. Then,*

$$\lim_{n\to\infty}\left[-\frac{\ln \bar{P}_{\rm FR}}{n}\right] = E_{\rm r}^{\rm FR}(R_{\rm w}). \tag{11}$$

Before providing the proof, a few points should be discussed.

1. First, observe that Theorem 1 asserts that $E_{\rm r}^{\rm FR}(R_{\rm w})$ is the *exact* random coding FR exponent, not just a lower bound. This is due to the fact that all steps of the analytic derivation are ensemble–tight in the exponential scale, thanks to the ability to avoid the use of the Jensen inequality and other well known tools that are traditionally used to facilitate the analysis, at the possible price of compromising tightness (see the proof of Theorem 1 below).

2. It is interesting to observe that the FR random coding exponent, $E_{\rm r}^{\rm FR}(R_{\rm w})$, depends only on $R_{\rm w}$, not on $R_{\rm s}$. This fact is not trivial, but the intuition is the following: in order to estimate $\boldsymbol{S}$ correctly, with high probability, from the given data $(\boldsymbol{Y}, \boldsymbol{W})$, there should be essentially no ambiguity, first of all, in defining what the correct $\boldsymbol{S}$ is. This will be the case if there is essentially only one source vector $\boldsymbol{X}$ that is responsible for the given $\boldsymbol{W}$ and then this $\boldsymbol{X}$ would dictate the correct $\boldsymbol{S} = g(\boldsymbol{X})$. This in turn would happen with high probability as long as $R_{\rm w} > H(X|Y)$. Otherwise, if more than one source vector (in the same conditional type class given $\boldsymbol{Y}$ as the correct one) is mapped by the encoder to the same helper message, then at least one such source vector is

likely to be mapped to a different secret key message, and then the decoding would be ambiguous. It appears then that correct estimation of $\boldsymbol{S}$ is essentially equivalent to correct estimation of $\boldsymbol{X}$, as in ordinary Slepian–Wolf decoding [6] (see also [15] and references therein), where there is no secret key at all (or alternatively, $R_{\mathrm{s}} \to \infty$). Indeed, the Slepian–Wolf coding component of the joint source–channel coding system, analyzed in [10, Section IV] under the generalized likelihood decoder, contributes the very same error exponent as asserted in Theorem 1.

3. It is interesting to examine a few decoding metrics. Consider the choice $a(Q) = -H_Q(X|Y)$. In this case, we have

$$\begin{aligned}
& \min_{Q_{X|Y}} [R_{\mathrm{w}} - H_Q(X|Y) + [a(Q_{X_0Y}) - a(Q_{XY})]_+]_+ \\
= \; & \min_{Q_{X|Y}} [R_{\mathrm{w}} - H_Q(X|Y) + [H_Q(X|Y) - H_Q(X_0|Y)]_+]_+ \\
= \; & \min_{Q_{X|Y}} [R_{\mathrm{w}} - \min\{H_Q(X|Y), H_Q(X_0|Y)\}]_+ \\
= \; & [R_{\mathrm{w}} - \min\{\max_{Q_{X|Y}} H_Q(X|Y), H_Q(X_0|Y)\}]_+ \\
= \; & [R_{\mathrm{w}} - H_Q(X_0|Y)]_+, \qquad (12)
\end{aligned}$$

which, together with (10), yields the same random coding exponent as the optimal MAP decoder for Slepian–Wolf decoding (see also [10] and [13]). More generally, the same comment applies to $a(Q) = -\beta H_Q(X|Y)$ for every $\beta \geq 1$, where $\beta \to \infty$ pertains to the deterministic universal minimum entropy decoding, the source–coding dual to maximum mutual information (MMI) universal decoding (see, e.g., [15] and references therein). For $a(Q) = \beta \boldsymbol{E}_Q \ln P(X|Y)$, we have a finite–temperature likelihood decoder. For $\beta \to \infty$, we are back to the ordinary MAP decoder, which yields

$$\begin{aligned}
& \lim_{\beta\to\infty} \min_{Q_{X|Y}} [R_{\mathrm{w}} - H_Q(X|Y) + [a(Q_{X_0Y}) - a(Q_{XY})]_+]_+ \\
= \; & \lim_{\beta\to\infty} \min_{Q_{X|Y}} [R_{\mathrm{w}} - H_Q(X|Y) + \beta[\boldsymbol{E}_Q \ln P(X_0|Y) - \boldsymbol{E}_Q \ln P(X|Y)]_+]_+ \\
= \; & \min_{\{Q_{X|Y}:\ \boldsymbol{E}_Q \ln P(X|Y) \geq \boldsymbol{E}_Q \ln P(X_0|Y)\}} [R_{\mathrm{w}} - H_Q(X|Y)]_+, \qquad (13)
\end{aligned}$$

which, together with (10), yields the random coding exponent of the MAP decoder, as expected. As argued above, this is the same as the exponent achieved by $a(Q) = -\beta H_Q(X|Y)$ for all $\beta \geq 1$.

The remaining part of this section is devoted to the proof of Theorem 1.

Proof of Theorem 1. The expected FR probability is given by

$$\bar{P}_{\rm FR} = \boldsymbol{E}\left\{\sum_{\boldsymbol{s}\neq\boldsymbol{S}} \tilde{P}(\boldsymbol{s}|\boldsymbol{W},\boldsymbol{Y})\right\} \tag{14}$$

where the expectation is w.r.t. both the randomness of $(\boldsymbol{S},\boldsymbol{W},\boldsymbol{Y})$ and the randomness of the code, $\mathcal{E}$. For given realizations, $\boldsymbol{X}=\boldsymbol{x}$ and $\boldsymbol{Y}=\boldsymbol{y}$, let us denote

$$\bar{P}_{\rm FR}(\boldsymbol{x},\boldsymbol{y}) \stackrel{\Delta}{=} \boldsymbol{E}\left\{\sum_{\boldsymbol{s}'\neq g(\boldsymbol{x})} \tilde{P}(\boldsymbol{s}'|f(\boldsymbol{x}),\boldsymbol{y})\right\}, \tag{15}$$

where now the expectation is merely w.r.t. the randomness of $\mathcal{E}$. Now, following eq. (3),

$$\begin{aligned}\tilde{P}(\boldsymbol{s}'|f(\boldsymbol{x}),\boldsymbol{y}) &= \frac{\sum_{\boldsymbol{x}'\in\mathcal{X}^n}\exp\{na(\hat{P}_{\boldsymbol{x}'\boldsymbol{y}})\}\cdot\mathcal{I}\{f(\boldsymbol{x}')=f(\boldsymbol{x})\}\cdot\mathcal{I}\{g(\boldsymbol{x}')=\boldsymbol{s}'\}}{\sum_{\boldsymbol{x}'\in\mathcal{X}^n}\exp\{na(\hat{P}_{\boldsymbol{x}'\boldsymbol{y}})\}\cdot\mathcal{I}\{f(\boldsymbol{x}')=f(\boldsymbol{x})\}}\\ &= \frac{\sum_{Q_{X|Y}} e^{na(Q_{XY})}N(\mathcal{T}(Q_{X|Y}|\boldsymbol{y}),f(\boldsymbol{x}),\boldsymbol{s}')}{e^{na(\hat{P}_{\boldsymbol{x}\boldsymbol{y}})}+\sum_{Q_{X|Y}} e^{na(Q_{XY})}N(\mathcal{T}(Q_{X|Y}|\boldsymbol{y}),f(\boldsymbol{x}))},\end{aligned} \tag{16}$$

where the summations over $\{Q_{X|Y}\}$ are across all conditional types $\{\mathcal{T}(Q_{X|Y}|\boldsymbol{y})\}$ of sequences of length n, and where

$$N(\mathcal{T}(Q_{X|Y}|\boldsymbol{y}),\boldsymbol{w},\boldsymbol{s}') = \left|\mathcal{T}(Q_{X|Y}|\boldsymbol{y})\bigcap\{\boldsymbol{x}':\ f(\boldsymbol{x}')=\boldsymbol{w},\ g(\boldsymbol{x}')=\boldsymbol{s}'\}\right|, \tag{17}$$

and

$$N(\mathcal{T}(Q_{X|Y}|\boldsymbol{y}),\boldsymbol{w}) = \left|\mathcal{T}(Q_{X|Y}\bigcap\{\boldsymbol{x}':\ f(\boldsymbol{x})=\boldsymbol{w},\ \boldsymbol{x}'\neq\boldsymbol{x}\}\right|. \tag{18}$$

Let us first consider the average FR probability for a given $(\boldsymbol{x},\boldsymbol{y})$ while fixing the realizations of $\boldsymbol{w}=f(\boldsymbol{x})$ and $\boldsymbol{s}=g(\boldsymbol{x})$:

$$\begin{aligned}\bar{P}_{\rm FR}(\boldsymbol{x},\boldsymbol{y},\boldsymbol{s},\boldsymbol{w}) &= \boldsymbol{E}\left\{\frac{\sum_{\boldsymbol{s}'\neq\boldsymbol{s}}\sum_{Q_{X|Y}} e^{na(Q_{XY})}N(\mathcal{T}(Q_{X|Y}|\boldsymbol{y}),f(\boldsymbol{x}),\boldsymbol{s}')}{e^{na(\hat{P}_{\boldsymbol{x}\boldsymbol{y}})}+\sum_{Q_{X|Y}} e^{na(Q_{XY})}N(\mathcal{T}(Q_{X|Y}|\boldsymbol{y}),f(\boldsymbol{x}))}\right\}\\ &= \int_0^1 {\rm d}t\cdot\Pr\left\{\frac{\sum_{Q_{X|Y}} e^{na(Q_{XY})}N(\mathcal{T}(Q_{X|Y}|\boldsymbol{y}),f(\boldsymbol{x}))}{e^{na(\hat{P}_{\boldsymbol{x}\boldsymbol{y}})}+\sum_{Q_{X|Y}} e^{na(Q_{XY})}N(\mathcal{T}(Q_{X|Y}|\boldsymbol{y}),f(\boldsymbol{x}))}\geq t\right\}\\ &= n\cdot\int_0^\infty {\rm d}\theta e^{-n\theta}\cdot\Pr\left\{\frac{\sum_{Q_{X|Y}} e^{na(Q_{XY})}N(\mathcal{T}(Q_{X|Y}|\boldsymbol{y}),f(\boldsymbol{x}))}{e^{na(\hat{P}_{\boldsymbol{x}\boldsymbol{y}})}+\sum_{Q_{X|Y}} e^{na(Q_{XY})}N(\mathcal{T}(Q_{X|Y}|\boldsymbol{y}),f(\boldsymbol{x}))}\geq e^{-n\theta}\right\}\end{aligned}$$

$$
\begin{aligned}
&\dot{=} \int_0^\infty \mathrm{d}\theta e^{-n\theta} \cdot \Pr\left\{ \sum_{Q_{X|Y}} e^{na(Q_{XY})} N(\mathcal{T}(Q_{X|Y}|\boldsymbol{y}), f(\boldsymbol{x})) > e^{n[a(\hat{P}_{\boldsymbol{xy}})-\theta]} \right\} \\
&\dot{=} \int_0^\infty \mathrm{d}\theta e^{-n\theta} \cdot \Pr\left\{ \max_{Q_{X|Y}} e^{na(Q_{XY})} N(\mathcal{T}(Q_{X|Y}|\boldsymbol{y}), f(\boldsymbol{x})) > e^{n[a(\hat{P}_{\boldsymbol{xy}})-\theta]} \right\} \\
&\dot{=} \int_0^\infty \mathrm{d}\theta e^{-n\theta} \cdot \Pr \bigcup_{Q_{X|Y}} \left\{ e^{na(Q_{XY})} N(\mathcal{T}(Q_{X|Y}|\boldsymbol{y}), f(\boldsymbol{x})) > e^{n[a(\hat{P}_{\boldsymbol{xy}})-\theta]} \right\} \\
&\dot{=} \max_{Q_{X|Y}} \int_0^\infty \mathrm{d}\theta e^{-n\theta} \cdot \Pr\left\{ N(\mathcal{T}(Q_{X|Y}|\boldsymbol{y}), f(\boldsymbol{x})) > e^{n[a(\hat{P}_{\boldsymbol{xy}})-a(Q_{XY})-\theta]} \right\}. \qquad (19)
\end{aligned}
$$

Now, observe that $N(\mathcal{T}(Q_{X|Y}|\boldsymbol{y}), f(\boldsymbol{x}))$ is a binomial random variable with $|\mathcal{T}(Q_{X|Y}|\boldsymbol{y})| \dot{=} e^{nH_Q(X|Y)}$ trials and probability of success $e^{-nR_{\mathrm{w}}}$. Similarly as argued, e.g., in [10] (see page 5042, bottom half of the right column therein), we have

$$
\Pr\left\{ N(\mathcal{T}(Q_{X|Y}|\boldsymbol{y}), f(\boldsymbol{x})) > e^{n[a(Q_{X_0Y})-a(Q_{XY})-\theta]} \right\} \dot{=} e^{-nE(Q_{XY}, Q_{X_0Y}, \theta, R_{\mathrm{w}})}, \qquad (20)
$$

where we have replaced $\hat{P}_{\boldsymbol{xy}}$ by the notation Q_{X_0Y} (X_0 being an auxiliary random variable that represents the underlying source vector $\boldsymbol{x}$), and where

$$
E(R_{\mathrm{w}}, Q_{X_0Y}, Q_{XY}, \theta) = \begin{cases} [R_{\mathrm{w}} - H_Q(X|Y)]_+ & \theta > a(Q_{X_0Y}) - a(Q_{XY}) - [H_Q(X|Y) - R_{\mathrm{w}}]_+ \\ \infty & \theta \le a(Q_{X_0Y}) - a(Q_{XY}) - [H_Q(X|Y) - R_{\mathrm{w}}]_+ \end{cases} \qquad (21)
$$

Thus,

$$
\bar{P}_{\mathrm{FR}}(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{s}, \boldsymbol{w}) \;\dot{=}\; \max_{Q_{X|Y}} \int_{[a(Q_{X_0Y})-a(Q_{XY})-[H_Q(X|Y)-R_{\mathrm{w}}]_+]_+}^{\infty} \mathrm{d}\theta e^{-n\theta} \cdot e^{-n[R_{\mathrm{w}}-H_Q(X|Y)]_+}. \qquad (22)
$$

whose exponential decay rate is according to

$$
\begin{aligned}
& \min_{Q_{X|Y}} \{ [a(Q_{X_0Y}) - a(Q_{XY}) - [H_Q(X|Y) - R_{\mathrm{w}}]_+]_+ + [R_{\mathrm{w}} - H_Q(X|Y)]_+ \} \\
= \; & \min_{Q_{X|Y}} \begin{cases} [R_{\mathrm{w}} - H_Q(X|Y) + a(Q_{X_0Y}) - a(Q_{XY})]_+ & H_Q(X|Y) > R_{\mathrm{w}} \\ R_{\mathrm{w}} - H_Q(X|Y) + [a(Q_{X_0Y}) - a(Q_{XY})]_+ & H_Q(X|Y) \le R_{\mathrm{w}} \end{cases} \\
= \; & \min_{Q_{X|Y}} [R_{\mathrm{w}} - H_Q(X|Y) + [a(Q_{X_0Y}) - a(Q_{XY})]_+]_+ \\
= \; & E(R_{\mathrm{w}}, Q_{X_0Y}). \qquad (23)
\end{aligned}
$$

The second to the last equality follows from the identity $[u - v]_+ = [[u]_+ - v]_+$, holding whenever $v \ge 0$, which is applied to the first line of the second expression with the assignments $u = a(Q_{X_0Y}) - a(Q_{XY})$ and $v = H_Q(X|Y) - R_{\mathrm{w}}$ (see also [13] as well as the text after eq. (11) of [10] for a very similar argument). Since this exponential behavior, of $\bar{P}_{\mathrm{FR}}(\boldsymbol{x}, \boldsymbol{y}, \boldsymbol{s}, \boldsymbol{w})$, is independent of the

particular realizations, $\boldsymbol{s}$ and $\boldsymbol{w}$, it holds also for the expectation w.r.t. the randomness of $\boldsymbol{S}$ and $\boldsymbol{W}$, namely, it also characterizes the exponential rate of $\bar{P}_{\mathrm{FR}}(\boldsymbol{x},\boldsymbol{y})$. Finally, it readily follows from the method of types [3] that the expectation w.r.t. the randomness of $(\boldsymbol{X},\boldsymbol{Y})$ decays according to the exponent

$$E_{\mathrm{r}}^{\mathrm{FR}}(R_{\mathrm{w}}) = \min_{Q_{X_0Y}} \{D(Q_{X_0Y}\|P_{XY}) + E(R_{\mathrm{w}}, Q_{X_0Y})\}, \tag{24}$$

which is as defined in (10). This completes the proof of Theorem 1. □

B. Expurgated Bound

Our expurgated bound will be asserted for each type class, $\mathcal{T}(Q_X)$, of source vectors separately. As in channel coding, where expurgation is associated with elimination of some 'bad' codewords of a randomly generated code, here too, we might need to eliminate a small fraction of bad source vectors from $\mathcal{T}(Q_X)$, in order to guarantee a certain FR performance level for each one of the remaining source vectors in $\mathcal{T}(Q_X)$. One may wonder what would be the justification for such an elimination of source vectors, as these are generated by the source and given to us, and they are not under our control. Nonetheless, in the context of biometric identification system described in Section III, where $\{\boldsymbol{x}\}$ are the enrollment signals, there are at least two possible ways to justify this elimination of a small fraction of the members of the type class.

1. In the enrollment stage, if the individual that subscribes to the system, has generated a 'forbidden' source vector $\boldsymbol{x}$ (in the sense that has been eliminated in the expurgation process), s/he might be asked to kindly provide his/her biometric signal once again, with the hope that this time a 'legitimate' source vector will be generated. The probability that this would happen is small in the first place, provided that the fraction of vectors eliminated from $\mathcal{T}(Q_X)$ is small. The probability of bothering the subscriber more than once with the request of a repeated measurement is even much smaller.

2. Considering the fact that $\boldsymbol{x}$ may be digitized with some precision (which is in line with the finite alphabet assumption anyway), it is conceivable to think of the enrollment data as having undergone a certain stage of vector quantization. Once $\boldsymbol{x}$ is thought of as an output of a vector quantizer, then not necessarily every member of $\mathcal{T}(Q_X)$ must be a legitimate codebook vector

in the first place. Among other things, one might rule out source vectors that contribute a high FR probability.

In order to present the expurgated exponent, a few additional definitions are needed. For a given Q_Y, let us define

$$\alpha(R_{\mathrm{w}}, Q_Y) \stackrel{\Delta}{=} \sup_{\{Q_{X|Y}:\ H_Q(X|Y)>R_{\mathrm{w}}\}} [a(Q_{XY}) + H_Q(X|Y)] - R_{\mathrm{w}}, \tag{25}$$

$$\gamma(Q_{XY}) \stackrel{\Delta}{=} \max\{a(Q_{XY}), \alpha(R_{\mathrm{w}}, Q_Y)\}, \tag{26}$$

$$\Lambda(Q_{XX'}) \stackrel{\Delta}{=} \min_{Q_{Y|XX'}} \{\gamma(Q_{XY}) - H_Q(Y|X, X') - \boldsymbol{E}_Q \ln P(Y|X) - a(Q_{X'Y})\}, \tag{27}$$

and for a given Q_X, define

$$E_{\mathrm{ex}}^{\mathrm{FR}}(R_{\mathrm{w}}, Q_X) = \inf_{\{Q_{X'|X}:\ H_Q(X'|X)\geq R_{\mathrm{w}}\}} \{\Lambda(Q_{XX'}) - H_Q(X'|X) + R_{\mathrm{w}}\}. \tag{28}$$

Finally, let $P_{\mathrm{FR}}(\mathcal{E}|\boldsymbol{x})$ denote the FR probability of a given enrollment encoder $\mathcal{E}$, conditioned on the input source vector $\boldsymbol{X} = \boldsymbol{x}$.

Theorem 2 *Consider the system configuration described in Section III and let $\{\delta_n\}_{n\geq 1}$ be a positive sequence tending to zero such that $n\delta_n \to \infty$. Then, there exists a code $\mathcal{E}$ such that for every Q_X,*

$$P_{\mathrm{FR}}(\mathcal{E}|\boldsymbol{x}) \leq \exp\{-nE_{\mathrm{ex}}^{\mathrm{FR}}(R_{\mathrm{w}}, Q_X) + o(n)\}, \tag{29}$$

for every $\boldsymbol{x} \in \mathcal{T}(Q_X) \setminus \mathcal{B}(Q_X)$, where $\mathcal{B}(Q_X)$ is a certain subset of $\mathcal{T}(Q_X)$, whose size does not exceed $e^{-n\delta_n}|\mathcal{T}(Q_X)|$.

A few points concerning Theorem 2 should be discussed.

1. It is interesting to note that the expression of $E_{\mathrm{ex}}^{\mathrm{FR}}(R_{\mathrm{w}}, Q_X)$ has some analogy to the Csiszár–Körner–Marton (CKM) expurgated exponent of channel coding [3, p. 165, Problem 10.18]. The term $\Lambda(Q_{XX'})$ plays the same role as the expected Bhattacharyya distance in the CKM expurgated exponent, whereas $H_Q(X'|X)$ is analogous to the coding rate R in channel coding and R_{w} is parallel to the empirical mutual information between channel codewords. Roughly speaking, the contribution of a single incorrect source vector $\boldsymbol{x}'$ to the FR probability is about $\exp\{-n\Lambda(Q_{XX'})$ provided that $(\boldsymbol{x}, \boldsymbol{x}') \in \mathcal{T}(Q_{XX'})$ (the pairwise error event). This probability should be multiplied

by the typical number of such incorrect source vectors within $\mathcal{T}(Q_{X'|X}|\boldsymbol{x})$ that are encoded into the same given helper message and hence may cause confusion. This number is of the exponential order $\exp\{n[H_Q(X'|X) - R_{\rm w}]\}$, provided that $H_Q(X'|X) - R_{\rm w} > 0$, and it vanishes otherwise.

2. Note that in contrast to Theorem 1, here we are no longer arguing that the result is ensemble–tight. There is actually one step in the derivation where exponential tightness might be compromised. Specifically, in one of the steps of this analysis, the denominator of (3) is lower bounded by a relatively simple single–letter bound that holds true for the vast majority of encoders, $\{\mathcal{E}\}$, in the ensemble. By doing this, possible gaps to these bounds may not be fully exploited, and we cannot rule out the possibility that this causes some loss of tightness. On the other hand, the derivation of the expurgated bound includes a certain degree of freedom that does not exist in the random coding bound of Theorem 1, and upon exploiting this degree of freedom, we obtain a result, which is at least as strong as the random coding bound, and sometimes strictly so.

3. The sequence δ_n tends to zero in order not to slow down the exponential decay rate, but it is also required that $n\delta_n \to \infty$ in order to guarantee that the set of 'bad' source vectors, $\mathcal{B}(Q_X)$, would be merely a minority of $\mathcal{T}(Q_X)$ for large n.

4. We now show that for every $R_{\rm w}$, the overall expurgated exponent (taking into account all types, $\{Q_X\}$) cannot be worse than $E_{\rm r}^{\rm FR}(R_{\rm w})$, at least for the metric $a(Q_{XY}) = -\beta H_Q(X|Y)$, which was shown to be as good as the optimal decoding metric in the ordinary random coding sense. Note that is in contrast to the traditional expurgated exponent, which improves on the random coding exponent only at a certain range of rates, but is inferior to the random coding exponent elsewhere (see also [10], where a similar finding was observed for a particular numerical example). For the above–mentioned choice of $a(Q_{XY})$, one easily verifies that $\alpha(R_{\rm w}, Q_Y) = -\beta R_{\rm w}$ and $\gamma(Q_{XY}) = -\beta\min\{H_Q(X|Y), R_{\rm w}\}$, and so,

$$\begin{aligned}\Lambda(Q_{XX'}) &= \min_{Q_{Y|XX'}}\{\gamma(Q_{XY}) - H_Q(Y|X,X') - \boldsymbol{E}_Q \ln P(Y|X) + \beta H_Q(X'|Y)\} \\ &= \min_{Q_{Y|XX'}}\{\beta[H_Q(X'|Y) - \min\{H_Q(X|Y), R_{\rm w}\}] + \\ &\quad I_Q(X';Y|X) + D(Q_{Y|X}\|P_{Y|X}|Q_X)\}. \end{aligned} \tag{30}$$

Upon optimizing β, we obtain

$$
\begin{aligned}
E_{\rm ex}(R_{\rm w},Q_X) &= \sup_{\beta\in\mathbb{R}}\inf_{\{Q_{X'|X}:\ H_Q(X'|X)\geq R_{\rm w}\}}\{\Lambda(Q_{XX'})-H_Q(X'|X)\}+R_{\rm w} \\
&= \sup_{\beta\in\mathbb{R}}\inf_{\{Q_{X'Y|X}:\ H_Q(X'|X)\geq R_{\rm w}\}}\{D(Q_{Y|X}\|P_{Y|X}|Q_X)+I_Q(X';Y|X)+ \\
&\quad \beta[H_Q(X'|Y)-\min\{H_Q(X|Y),R_{\rm w}\}]-H_Q(X'|X)+R_{\rm w}\} \\
&\geq \inf_{\{Q_{X'Y|X}:\ H_Q(X'|X)\geq R_{\rm w}\}}\{D(Q_{Y|X}\|P_{Y|X}|Q_X)+I_Q(X';Y|X)+ \\
&\quad H_Q(X'|Y)-\min\{H_Q(X|Y),R_{\rm w}\}-H_Q(X'|X)+R_{\rm w}\} \\
&= \inf_{\{Q_{X'Y|X}:\ H_Q(X'|X)\geq R_{\rm w}\}}\{D(Q_{Y|X}\|P_{Y|X}|Q_X)+I_Q(X';Y|X)+H_Q(X'|Y)+ \\
&\quad [R_{\rm w}-H_Q(X|Y)]_+-H_Q(X'|X) \\
&= \inf_{\{Q_{X'Y|X}:\ H_Q(X'|X)\geq R_{\rm w}\}}\{D(Q_{Y|X}\|P_{Y|X}|Q_X)+H_Q(X'|Y)-H_Q(X'|X,Y)+ \\
&\quad [R_{\rm w}-H_Q(X|Y)]_+ \\
&= \inf_{\{Q_{X'Y|X}:\ H_Q(X'|X)\geq R_{\rm w}\}}\{D(Q_{Y|X}\|P_{Y|X}|Q_X)+I_Q(X';X|Y)+[R_{\rm w}-H_Q(X|Y)]_+\} \\
&\geq \inf_{\{Q_{X'Y|X}:\ H_Q(X'|X)\geq R_{\rm w}\}}\{D(Q_{Y|X}\|P_{Y|X}|Q_X)+[R_{\rm w}-H_Q(X|Y)]_+\}. \qquad (31)
\end{aligned}
$$

Without the constraint, $H_Q(X'|X)\geq R_{\rm w}$, the last expression is exactly the random coding FR exponent for a given type Q_X, and upon taking into account the probabilistic weight of each type, the overall exponent associated with the last line (again, without the constraint) is exactly $E_{\rm r}(R_{\rm m})$ of Theorem 1 for the optimal, MAP decoder. By inspection of eq. (31), we therefore observe that there are four origins of the gap between the expurgated exponent and the random coding exponent: (i) the decoder actually being analyzed might be suboptimal for the expurgated ensemble, (ii) the optimal β (for the given family of decoders) might not necessarily be $\beta^*=1$ (the first inequality in the above chain). In fact, the optimal β^* is expected to depend on $R_{\rm w}$.[1] (iii) the term $I_Q(X';X|Y)$ which may not necessarily vanish for the optimal $Q_{X'Y|X}$ (the second inequality), and (iv) the constraint $H_Q(X'|X)\geq R_{\rm w}$. For example, if $R_{\rm w}>\ln|\mathcal{X}|$, the expurgated exponent is infinite while the random coding exponent is finite.

5. As can be seen in the proof of Theorem 2, the asserted expurgated exponent is obtained from

[1] The fact that optimal β may not necessarily be infinite (except the case (5)), is interesting on its own right, as it means that the the stochastic decoder may outperform the deterministic one for a given (suboptimal) decoding metric.

an intermediate expression that depends on a free parameter ρ that undergoes optimization. It is interesting to observe what happens when we set $\rho = 1$ instead of optimizing over ρ. This would correspond to the ordinary ensemble average, which needs no expurgation. In this case, $E_{\rm ex}^{\rm FR}(R_{\rm w}, Q_X)$ would be replaced by

$$\begin{aligned} E_1(R_{\rm w}, Q_X) &= \sup_{\beta\in\mathbb{R}} \inf_{Q_{X'|X}} \{\Lambda(Q_{XX'}) - [H_Q(X'|X) - R_{\rm w}]_+ + [R_{\rm w} - H_Q(X'|X)]_+\} \\ &= \sup_{\beta\in\mathbb{R}} \inf_{Q_{X'|X}} \{\Lambda(Q_{XX'}) + R_{\rm w} - H_Q(X'|X)\}\,, \end{aligned} \tag{32}$$

where we have used the trivial identity $[u]_+ - [-u]_+ \equiv u$. Therefore, the expression of $E_1(R_{\rm w}, Q_X)$ is exactly like that of $E_{\rm ex}^{\rm FR}(R_{\rm m}, Q_X)$, except that the constraint, $H_Q(X'|X) \geq R_{\rm w}$, is removed. It follows that $E_{\rm ex}^{\rm FR}(R_{\rm w}, Q_X)$ is expected to improve on $E_1(R_{\rm w}, Q_X)$ at high rates, where the constraint may be active. It also follows (similarly as in (31)) that $E_1(R_{\rm w}, Q_X)$ is never smaller than the random coding FR exponent given the type Q_X, since the latter lacks this constraint as well. The reason that this expurgated exponent is nowhere worse than the random coding exponent is that we do not use the inequality $[\sum_{\boldsymbol{x}'\neq\boldsymbol{x}} u(\boldsymbol{x}')]^{1/\rho} \leq \sum_{\boldsymbol{x}'\neq\boldsymbol{x}} [u(\boldsymbol{x}')]^{1/\rho}$ (holding for $\rho \geq 1$), like in the traditional expurgated bound. This inequality causes a loss of tightness. Without it, the supremum over ρ is always achieved at $\rho \to \infty$.

6. The case of ordinary, deterministic MAP decoding is obtained again as of special case of (5) in the limit $\beta \to \infty$. As in (13), when the objective function to be minimized over $\{Q_{XX'Y})\}$, contains a term like $\beta \cdot G(Q_{XX'Y})$ (for some functional $G(\cdot)$), then in the limit of $\beta \to \infty$, it is replaced by a constraint of the form $G(Q_{XX'Y}) \leq 0$.

The remaining part of this section is devoted to the proof of Theorem 2.

Proof of Theorem 2. For a given code, $\mathcal{E}$, and a given the underlying source vector $\boldsymbol{x}$, we have

$$P_{\rm FR}(\mathcal{E}|\boldsymbol{x}) = \sum_{\boldsymbol{y}} P(\boldsymbol{y}|\boldsymbol{x}) \sum_{\boldsymbol{s}\neq g(\boldsymbol{x})} \tilde{P}(\boldsymbol{s}|f(\boldsymbol{x}), \boldsymbol{y}) \tag{33}$$

$$= \sum_{\boldsymbol{s}\neq g(\boldsymbol{x})} \sum_{\boldsymbol{y}} P(\boldsymbol{y}|\boldsymbol{x}) \cdot \frac{\tilde{P}(\boldsymbol{s}, f(\boldsymbol{x})|\boldsymbol{y})}{\exp\{na(\hat{P}_{\boldsymbol{x}\boldsymbol{y}})\} + Z_{\boldsymbol{x}}(\boldsymbol{y})}, \tag{34}$$

where

$$Z_{\boldsymbol{x}}(\boldsymbol{y}) = \sum_{\boldsymbol{x}'\neq\boldsymbol{x}} \exp\{na(\hat{P}_{\boldsymbol{x}'\boldsymbol{y}})\} \cdot \mathcal{I}\{f(\boldsymbol{x}') = f(\boldsymbol{x})\}. \tag{35}$$

Let $\epsilon > 0$ be arbitrarily small. It is shown in the Appendix[2] that

$$\Pr\Big\{Z_{\boldsymbol{x}}(\boldsymbol{y}) < \exp\{n\alpha(R_{\rm w}+\epsilon, \hat{P}_{\boldsymbol{y}})\} \text{ for some } (\boldsymbol{x},\boldsymbol{y})\Big\} \le |\mathcal{X}\times\mathcal{Y}|^n \cdot \exp\{-e^{n\epsilon}+n\epsilon+1\}. \tag{36}$$

Now, denoting

$$\mathcal{G}_\epsilon = \Big\{\mathcal{E}:\ Z_{\boldsymbol{x}}(\boldsymbol{y}) \ge \exp\{n\alpha(R_{\rm w}+\epsilon, \hat{P}_{\boldsymbol{y}})\} \text{ for all } (\boldsymbol{x},\boldsymbol{y})\Big\}, \tag{37}$$

we have:

$$\begin{aligned}
& \boldsymbol{E}\left\{[P_{\rm FR}(\mathcal{E}|\boldsymbol{x})]^{1/\rho}\right\} \\
= \ & \boldsymbol{E}\left[\sum_{\boldsymbol{s}\neq g(\boldsymbol{x})}\sum_{\boldsymbol{y}} P(\boldsymbol{y}|\boldsymbol{x})\cdot\frac{\tilde{P}(\boldsymbol{s},f(\boldsymbol{x})|\boldsymbol{y})}{\exp\{na(\hat{P}_{\boldsymbol{xy}})\}+Z_{\boldsymbol{x}}(\boldsymbol{y})}\right]^{1/\rho} \\
= \ & \sum_{\mathcal{E}} P(\mathcal{E})\left[\sum_{\boldsymbol{s}\neq g(\boldsymbol{x})}\sum_{\boldsymbol{y}} P(\boldsymbol{y}|\boldsymbol{x})\cdot\frac{\tilde{P}(\boldsymbol{s},f(\boldsymbol{x})|\boldsymbol{y})}{\exp\{na(\hat{P}_{\boldsymbol{xy}})\}+Z_{\boldsymbol{x}}(\boldsymbol{y})}\right]^{1/\rho} \\
= \ & \sum_{\mathcal{E}\in\mathcal{G}_\epsilon} P(\mathcal{E})\left[\sum_{\boldsymbol{s}\neq g(\boldsymbol{x})}\sum_{\boldsymbol{y}} P(\boldsymbol{y}|\boldsymbol{x})\cdot\frac{\tilde{P}(\boldsymbol{s},f(\boldsymbol{x})|\boldsymbol{y})}{\exp\{na(\hat{P}_{\boldsymbol{xy}})\}+Z_{\boldsymbol{x}}(\boldsymbol{y})}\right]^{1/\rho} + \\
& \sum_{\mathcal{E}\in\mathcal{G}_\epsilon^c} P(\mathcal{E})\left[\sum_{\boldsymbol{s}\neq g(\boldsymbol{x})}\sum_{\boldsymbol{y}} P(\boldsymbol{y}|\boldsymbol{x})\cdot\frac{\tilde{P}(\boldsymbol{s},f(\boldsymbol{x})|\boldsymbol{y})}{\exp\{na(\hat{P}_{\boldsymbol{xy}})\}+Z_{\boldsymbol{x}}(\boldsymbol{y})}\right]^{1/\rho} \\
\le \ & \sum_{\mathcal{E}\in\mathcal{G}_\epsilon} P(\mathcal{E})\left[\sum_{\boldsymbol{s}\neq g(\boldsymbol{x})}\sum_{\boldsymbol{y}} P(\boldsymbol{y}|\boldsymbol{x})\cdot\frac{\tilde{P}(\boldsymbol{s},f(\boldsymbol{x})|\boldsymbol{y})}{\exp\{na(\hat{P}_{\boldsymbol{xy}})\}+\exp\{n\alpha(R_{\rm w}+\epsilon,\hat{P}_{\boldsymbol{y}})\}}\right]^{1/\rho} + \\
& \sum_{\mathcal{E}\in\mathcal{G}_\epsilon^c} P(\mathcal{E})\cdot 1^{1/\rho} \\
\le \ & \sum_{\mathcal{E}} P(\mathcal{E})\left[\sum_{\boldsymbol{s}\neq g(\boldsymbol{x})}\sum_{\boldsymbol{y}} P(\boldsymbol{y}|\boldsymbol{x})\cdot\frac{\tilde{P}(\boldsymbol{s},f(\boldsymbol{x})|\boldsymbol{y})}{\exp\{na(\hat{P}_{\boldsymbol{xy}})\}+\exp\{n\alpha(R_{\rm w}+\epsilon,\hat{P}_{\boldsymbol{y}})\}}\right]^{1/\rho} + \\
& e^{nR_{\rm s}}\cdot|\mathcal{X}\times\mathcal{Y}|^n\cdot\exp\{-e^{n\epsilon}+n\epsilon+1\}.
\end{aligned} \tag{38}$$

Considering the arbitrariness of ϵ, the expression in the square brackets is exponentially equivalent to

$$\begin{aligned}
& \sum_{\boldsymbol{s}\neq g(\boldsymbol{x})}\sum_{\boldsymbol{y}} P(\boldsymbol{y}|\boldsymbol{x})e^{-n\gamma(\hat{P}_{\boldsymbol{xy}})}\tilde{P}(\boldsymbol{s},f(\boldsymbol{x})|\boldsymbol{y}) \\
= \ & \sum_{\boldsymbol{s}\neq g(\boldsymbol{x})}\sum_{\boldsymbol{y}} P(\boldsymbol{y}|\boldsymbol{x})e^{-n\gamma(\hat{P}_{\boldsymbol{xy}})}\sum_{\boldsymbol{x}'}\exp\{na(\hat{P}_{\boldsymbol{x}'\boldsymbol{y}})\}\mathcal{I}\{f(\boldsymbol{x}')=f(\boldsymbol{x}),g(\boldsymbol{x}')=\boldsymbol{s}\} \\
= \ & \sum_{\boldsymbol{s}\neq g(\boldsymbol{x})}\sum_{\boldsymbol{x}'}\mathcal{I}\{f(\boldsymbol{x}')=f(\boldsymbol{x}),g(\boldsymbol{x}')=\boldsymbol{s}\}\sum_{\boldsymbol{y}} P(\boldsymbol{y}|\boldsymbol{x})\exp\{n[a(\hat{P}_{\boldsymbol{x}'\boldsymbol{y}})-\gamma(\hat{P}_{\boldsymbol{xy}})]\}.
\end{aligned} \tag{39}$$

[2]See also [10, Appendix B] for a similar argument related to channel coding.

Now, the inner most summation (over $\boldsymbol{y}$) can be assessed using the method of types [3]. Accordingly, referring to (27), we have

$$e^{-n\Lambda(\hat{P}_{\boldsymbol{x}\boldsymbol{x}'})} \doteq \sum_{\boldsymbol{y}} P(\boldsymbol{y}|\boldsymbol{x}) \exp\{n[a(\hat{P}_{\boldsymbol{x}'\boldsymbol{y}}) - \gamma(\hat{P}_{\boldsymbol{x}\boldsymbol{y}})]\}, \tag{40}$$

which is the contribution of a single incorrect source vector $\boldsymbol{x}'$ to the FR probability. This yields

$$\begin{aligned} & \sum_{\boldsymbol{s}\neq g(\boldsymbol{x})} \sum_{\boldsymbol{x}'} \mathcal{I}\{f(\boldsymbol{x}') = f(\boldsymbol{x}), g(\boldsymbol{x}') = \boldsymbol{s}\} \cdot e^{-n\Lambda(\hat{P}_{\boldsymbol{x}\boldsymbol{x}'})} \\ \leq & \sum_{\boldsymbol{x}'} e^{-n\Lambda(\hat{P}_{\boldsymbol{x}\boldsymbol{x}'})} \mathcal{I}\{f(\boldsymbol{x}') = f(\boldsymbol{x})\} \\ = & \sum_{Q_{X'|X}} e^{-n\Lambda(Q_{XX'})} N(\mathcal{T}(Q_{X'|X}|\boldsymbol{x}), f(\boldsymbol{x})), \end{aligned} \tag{41}$$

where we have defined

$$N(\mathcal{T}(Q_{X'|X}|\boldsymbol{x}), f(\boldsymbol{x})) \triangleq \Big|\mathcal{T}(Q_{X'|X}|\boldsymbol{x}) \cap \{\boldsymbol{x}' : \ f(\boldsymbol{x}') = f(\boldsymbol{x})\}\Big|. \tag{42}$$

On substituting this back into the bound on $\boldsymbol{E}\left\{[P_{\mathrm{FR}}(\mathcal{E}|\boldsymbol{x})]^{1/\rho}\right\}$, we get

$$\begin{aligned} & \boldsymbol{E}\left\{[P_{\mathrm{e}}(\mathcal{E}|\boldsymbol{x})]^{1/\rho}\right\} \\ \leq & \boldsymbol{E}\left\{\left[\sum_{Q_{X'|X}} e^{-n\Lambda(Q_{XX'})} N(\mathcal{T}(Q_{X'|X}|\boldsymbol{x}), f(\boldsymbol{x}))\right]^{1/\rho}\right\} \\ \doteq & \sum_{Q_{X'|X}} e^{-n\Lambda(Q_{XX'})/\rho} \boldsymbol{E}\left\{[N(\mathcal{T}(Q_{X'|X}|\boldsymbol{x}), f(\boldsymbol{x}))]^{1/\rho}\right\} \\ = & \sum_{Q_{X'|X}} e^{-n\Lambda(Q_{XX'})/\rho} \int_0^\infty \mathrm{d}t \cdot \Pr\left\{\left[N(\mathcal{T}(Q_{X'|X}|\boldsymbol{x}), f(\boldsymbol{x}))\right]^{1/\rho} \geq t\right\} \\ = & \sum_{Q_{X'|X}} e^{-n\Lambda(Q_{XX'})/\rho} \int_0^\infty \mathrm{d}t \cdot \Pr\left\{N(\mathcal{T}(Q_{X'|X}|\boldsymbol{x}), f(\boldsymbol{x})) \geq t^\rho\right\} \\ \doteq & \sum_{Q_{X'|X}} e^{-n\Lambda(Q_{XX'})/\rho} \int_{-\infty}^\infty \mathrm{d}\theta \cdot e^{n\theta} \cdot \Pr\left\{N(\mathcal{T}(Q_{X'|X}|\boldsymbol{x}), f(\boldsymbol{x})) \geq e^{n\theta\rho}\right\}. \end{aligned} \tag{43}$$

Let us focus on the term $\Pr[N(\mathcal{T}(Q_{X'|X}|\boldsymbol{x}), f(\boldsymbol{x})) \geq e^{n\theta\rho}]$. Since $N(\mathcal{T}(Q_{X'|X}|\boldsymbol{x}), f(\boldsymbol{x}))$ is a binomial random variable with $|\mathcal{T}(Q_{X'|X}|\boldsymbol{x})| \doteq e^{nH_Q(X'|X)}$ trials and probability of success $e^{-nR_{\mathrm{w}}}$, we have

$$\Pr\left[N(\mathcal{T}(Q_{X'|X}|\boldsymbol{x}), f(\boldsymbol{x})) \geq e^{n\theta\rho}\right] \doteq e^{-nE(R_{\mathrm{w}}, Q_{XX'}, \rho\theta)} \tag{44}$$

where

$$E(R_{\rm w}, Q_{XX'}, \rho\theta) = \begin{cases} [R_{\rm w} - H_Q(X'|X)]_+ & [H_Q(X'|X) - R_{\rm w}]_+ \geq \rho\theta \\ \infty & [H_Q(X'|X) - R_{\rm w}]_+ < \rho\theta \end{cases}$$
$$= \begin{cases} [R_{\rm w} - H_Q(X'|X)]_+ & \theta \leq [H_Q(X'|X) - R_{\rm w}]_+/\rho \\ \infty & \theta > [H_Q(X'|X) - R_{\rm w}]_+/\rho \end{cases} \quad (45)$$

On substituting this back into the expression of $\boldsymbol{E}\left\{[P_{\rm FR}(\mathcal{E}|\boldsymbol{x})]^{1/\rho}\right\}$, we get

$$\begin{aligned} & \boldsymbol{E}\left\{[P_{\rm FR}(\mathcal{E}|\boldsymbol{x})]^{1/\rho}\right\} \\ \dot{\leq} & \sum_{Q_{X'|X}} e^{-n\Lambda(Q_{XX'})/\rho} \cdot \int_{-\infty}^{[H_Q(X'|X)-R_{\rm w}]_+/\rho} d\theta \cdot e^{n\theta} e^{-n[R_{\rm w}-H_Q(X'|X)]_+} \\ \dot{=} & \exp\left\{-n \min_{Q_{X'|X}} \left[\Lambda(Q_{XX'}) + \rho[R_{\rm w} - H_Q(X'|X)]_+ - [H_Q(X'|X) - R_{\rm w}]_+\right]/\rho\right\} \\ \stackrel{\Delta}{=} & e^{-nE_{\rm x}(R_{\rm w},Q_X,\rho)/\rho}. \end{aligned} \quad (46)$$

It follows then that

$$\boldsymbol{E}\left\{\frac{1}{|\mathcal{T}(Q_X)|}\sum_{\boldsymbol{x}\in\mathcal{T}(Q_X)}[P_{\rm FR}(\mathcal{E}|\boldsymbol{x})]^{1/\rho}\right\} \dot{\leq} e^{-nE_{\rm x}(R_{\rm w},Q_X,\rho)/\rho}, \quad (47)$$

and so, there exists a code $\mathcal{E}$ with

$$\frac{1}{|\mathcal{T}(Q_X)|}\sum_{\boldsymbol{x}\in\mathcal{T}(Q_X)}[P_{\rm FR}(\mathcal{E}|\boldsymbol{x})]^{1/\rho} \dot{\leq} e^{-nE_{\rm x}(R_{\rm w},Q_X,\rho)/\rho}. \quad (48)$$

For a given such $\mathcal{E}$ and Q_X, let us order the members of $\mathcal{T}(Q_X)$, as $\boldsymbol{x}_1, \boldsymbol{x}_2, \boldsymbol{x}_3, \ldots$, according to $P_{\rm FR}(\mathcal{E}|\boldsymbol{x}_1) \geq P_{\rm FR}(\mathcal{E}|\boldsymbol{x}_2) \geq P_{\rm FR}(\mathcal{E}|\boldsymbol{x}_3) \geq \ldots$ and let M be a temporary short-hand notation for $|\mathcal{T}(Q_X)|$. Let $\mathcal{B}(Q_X)$ be the subset of $\mathcal{T}(Q_X)$ formed by the first $M' = e^{-\delta n}M$ members of $\mathcal{T}(Q_X)$ according to this order, i.e., $B(Q_X) = \{\boldsymbol{x}_1, \boldsymbol{x}_2, \ldots, \boldsymbol{x}_{M'}\}$. We then have

$$\begin{aligned} e^{-nE_{\rm x}(R_{\rm w},Q_X,\rho)/\rho} & \dot{\geq} \frac{1}{M}\sum_{m=1}^{M}[P_{\rm FR}(\mathcal{E}|\boldsymbol{x}_m)]^{1/\rho} \\ & \geq \frac{1}{M}\sum_{m=1}^{M'}[P_{\rm FR}(\mathcal{E}|\boldsymbol{x}_m)]^{1/\rho} \\ & \geq \frac{1}{M}\sum_{m=1}^{M'}[P_{\rm FR}(\mathcal{E}|\boldsymbol{x}_{M'+1})]^{1/\rho} \\ & = \frac{1}{M}\cdot M' \cdot [P_{\rm FR}(\mathcal{E}|\boldsymbol{x}_{M'+1})]^{1/\rho} \\ & = e^{-n\delta_n}\left[\max_{\boldsymbol{x}\in\mathcal{T}(Q_X)\setminus B(Q_X)} P_{\rm FR}(\mathcal{E}|\boldsymbol{x})\right]^{1/\rho}, \end{aligned} \quad (49)$$

and so, $\max_{\boldsymbol{x}\in\mathcal{T}(Q_X)\setminus B(Q_X)} P_{\rm FR}(\mathcal{E}|\boldsymbol{x})$ decays at an exponential rate which is at least as large as

$$\begin{aligned}
& \sup_{\rho\geq 0} E_{\rm x}(R_{\rm w}, Q_X, \rho) \\
= & \sup_{\rho\geq 0}\inf_{Q_{X'|X}} \{\Lambda(Q_{XX'}) - [H_Q(X'|X) - R_{\rm w}]_+ + \rho[R_{\rm w} - H_Q(X'|X)]_+\} \qquad (50)\\
= & \inf_{\{Q_{X'|X}:\ H_Q(X'|X)\geq R_{\rm w}\}} \{\Lambda(Q_{XX'}) - [H_Q(X'|X) - R_{\rm w}]_+\} \\
= & \inf_{\{Q_{X'|X}:\ H_Q(X'|X)\geq R_{\rm w}\}} \{\Lambda(Q_{XX'}) - H_Q(X'|X) + R_{\rm w}\} \\
= & E_{\rm x}(R_{\rm w}, Q_X), \qquad (51)
\end{aligned}$$

completing the proof of Theorem 2.

V. False–Accept Error Analysis

In this section, we analyze the ensemble performance of the system from the viewpoint of an imposter who makes an attempt to estimate the secret key without access to the side information $\boldsymbol{Y}$, and we are interested in the exponential decay rate of the FA probability for the average code. As described in Section III, here we assume that the imposter estimates $\boldsymbol{S}$ using the MAP estimator, $\tilde{\boldsymbol{S}}$ (see (8)), based on the helper message only. Accordingly, as defined in Section III, we denote $\bar{P}_{\rm FA} = \Pr\{\tilde{\boldsymbol{S}} = \boldsymbol{S}\}$, i.e., the probability of correct decoding (FA), averaged over the ensemble of codes $\{\mathcal{E}\}$. Let us define

$$E_{\rm FA}(R_{\rm w}, R_{\rm s}) = \min_{Q_X} \left[D(Q_X\|P_X) + \min\{R_{\rm s}, [H_Q(X) - R_{\rm w}]_+\}\right]. \qquad (52)$$

Our main result, in this section, is the following.

Theorem 3 *Consider the system configuration described in Section III. Then,*

$$\bar{P}_{\rm FA} \leq \exp\{-nE_{\rm FA}(R_{\rm w}, R_{\rm s}) + o(n)\}. \qquad (53)$$

The expression of this exponential error bound is quite intuitive and it can easily be understood to hold even if the imposter is informed about the type[3] Q_X of $\boldsymbol{X}$. There are about $e^{n[H_Q(X)-R_{\rm w}]_+}$ source sequences of type Q_X (including the correct one), whose helper message is the given $\boldsymbol{W}$. If

[3]Here a genie–aided decoding argument does not harm the tightness of the FA exponent, because one can guess the type correctly with probability of success that decays only polynomially.

$[H_Q(X) - R_{\rm w}]_+ > R_{\rm s}$, then all possible $e^{nR_{\rm s}}$ members of the secret–message set would be likely to appear as encoded secret messages among those sequences, approximately evenly, so the probability of guessing the correct one is about $e^{-nR_{\rm s}}$. If, on the other hand, $[H_Q(X) - R_{\rm w}]_+ < R_{\rm s}$, then it is very likely that there would be only about $e^{n[H_Q(X)-R_{\rm w}]_+}$ different $\boldsymbol{s}$–messages, so the probability of guessing the correct one is the reciprocal, $e^{-n[H_Q(X)-R_{\rm w}]_+}$. It is easy to see that $E_{\rm FA}(R_{\rm w}, R_{\rm s})$ vanishes for $R_{\rm w} > H(X)$, as expected.

It is also interesting to observe that here, in contrast to the exponential FR bounds of Section IV, the exponent depends on both $R_{\rm w}$ and $R_{\rm s}$, and not only on $R_{\rm w}$. As expected, it is increasing in $R_{\rm s}$ and decreasing in $R_{\rm w}$.

The FA error exponent of Theorem 3 can also be presented in a Gallager–style form:

$$\begin{aligned}
E_{\rm FA}(R_{\rm w}, R_{\rm s}) &= \min_{Q}[D(Q_X\|P_X) + \min\{R_{\rm s}, [H_Q(X) - R_{\rm w}]_+\}] \\
&= \min_{Q_X}\min_{0\le s\le 1}\max_{0\le\rho\le 1}\{D(Q_X\|P_X) + sR_{\rm s} + (1-s)\rho[H_Q(X) - R_{\rm w}]\} \\
&= \min_{0\le s\le 1}\max_{0\le\rho\le 1}\min_{Q_X}\{D(Q_X\|P_X) + sR_{\rm s} + (1-s)\rho[H_Q(X) - R_{\rm w}]\} \\
&= \min_{0\le s\le 1}\max_{0\le\rho\le 1}\left\{-[1-\rho(1-s)]\ln\left[\sum_x P_X(x)^{1/[1-\rho(1-s)]}\right] + sR_{\rm s} - \rho(1-s)R_{\rm w}\right\} \\
&= \min_{0\le s\le 1}\max_{s\le\rho\le 1}\left\{-\rho\ln\left[\sum_x P_X(x)^{1/\rho}\right] + sR_{\rm s} - (1-\rho)R_{\rm w}\right\}. \qquad (54)
\end{aligned}$$

Proof of Theorem 3. In the derivation below, we let $\boldsymbol{x}_Q$ denote an arbitrary representative source vector $\boldsymbol{x}$ of type Q_X. The choice of this representative within $\mathcal{T}(Q_X)$ is completely immaterial since all members of $\mathcal{T}(Q_X)$ are equiprobable. Similarly as before, we also denote by $N(Q_X, \boldsymbol{w}, \boldsymbol{s})$ the number of members of $\mathcal{T}(Q_X)$ that are encoded into $(\boldsymbol{w}, \boldsymbol{s})$.

$$\begin{aligned}
\bar{P}_{\rm FA} &= \boldsymbol{E}\left\{\sum_{\boldsymbol{w}}\max_{\boldsymbol{s}} P(\boldsymbol{w}, \boldsymbol{s})\right\} \\
&= \boldsymbol{E}\left\{\sum_{\boldsymbol{w}}\max_{\boldsymbol{s}}\sum_{Q_X} P_X(\boldsymbol{x}_Q)\cdot N(Q_X, \boldsymbol{w}, \boldsymbol{s})\right\} \\
&= \lim_{\beta\to\infty}\boldsymbol{E}\left\{\sum_{\boldsymbol{w}}\left[\sum_{\boldsymbol{s}}\left(\sum_{Q_X} P_X(\boldsymbol{x}_Q)\cdot N(Q_X, \boldsymbol{w}, \boldsymbol{s})\right)^{\beta}\right]^{1/\beta}\right\} \\
&\doteq \lim_{\beta\to\infty}\boldsymbol{E}\left\{\sum_{\boldsymbol{w}}\left[\sum_{\boldsymbol{s}}\sum_{Q_X} P_X^{\beta}(\boldsymbol{x}_Q)\cdot N^{\beta}(Q_X, \boldsymbol{w}, \boldsymbol{s})\right]^{1/\beta}\right\}
\end{aligned}$$

$$
\begin{aligned}
&= \lim_{\beta\to\infty} \boldsymbol{E}\left\{\sum_{\boldsymbol{w}}\left[\sum_{Q_X}\sum_{\boldsymbol{s}} P_X^{\beta}(\boldsymbol{x}_Q)\cdot N^{\beta}(Q_X,\boldsymbol{w},\boldsymbol{s})\right]^{1/\beta}\right\}\\
&\dot{=} \lim_{\beta\to\infty} \boldsymbol{E}\left\{\sum_{\boldsymbol{w}}\sum_{Q_X}\left[\sum_{\boldsymbol{s}} P_X^{\beta}(\boldsymbol{x}_Q)\cdot N^{\beta}(Q_X,\boldsymbol{w},\boldsymbol{s})\right]^{1/\beta}\right\}\\
&= \lim_{\beta\to\infty} \boldsymbol{E}\left\{\sum_{\boldsymbol{w}}\sum_{Q_X} P_X(\boldsymbol{x}_Q)\left[\sum_{\boldsymbol{s}} N^{\beta}(Q_X,\boldsymbol{w},\boldsymbol{s})\right]^{1/\beta}\right\}\\
&= \lim_{\beta\to\infty} \sum_{\boldsymbol{w}}\sum_{Q_X} P_X(\boldsymbol{x}_Q)\cdot \boldsymbol{E}\left\{\left[\sum_{\boldsymbol{s}} N^{\beta}(Q_X,\boldsymbol{w},\boldsymbol{s})\right]^{1/\beta}\right\}\\
&= \sum_{\boldsymbol{w}}\sum_{Q_X} P_X(\boldsymbol{x}_Q)\cdot \boldsymbol{E}\left\{\max_{\boldsymbol{s}} N(Q_X,\boldsymbol{w},\boldsymbol{s})\right\}\\
&= \sum_{\boldsymbol{w}}\sum_{Q_X} P_X(\boldsymbol{x}_Q)\cdot \sum_{n=1}^{|\mathcal{T}(Q_X)|}\Pr\left\{\max_{\boldsymbol{s}} N(Q_X,\boldsymbol{w},\boldsymbol{s})\geq n\right\}\\
&= \sum_{\boldsymbol{w}}\sum_{Q_X} P_X(\boldsymbol{x}_Q)\cdot \sum_{n=1}^{|\mathcal{T}(Q_X)|}\Pr\bigcup_{\boldsymbol{s}}\{N(Q_X,\boldsymbol{w},\boldsymbol{s})\geq n\}\\
&\leq \sum_{\boldsymbol{w}}\sum_{Q_X} P_X(\boldsymbol{x}_Q)\cdot \sum_{n=1}^{|\mathcal{T}(Q_X)|}\min\left\{1, e^{nR_{\mathrm{s}}}\Pr\left[N(Q_X,\boldsymbol{w},\boldsymbol{s})\geq n\right]\right\}. \qquad (55)
\end{aligned}
$$

Now, for $Q_X\in\mathcal{G}\stackrel{\Delta}{=}\{Q_X:\ H_Q(X)>R_{\mathrm{s}}+R_{\mathrm{w}}\}$, clearly, $\Pr[N(Q_X,\boldsymbol{w},\boldsymbol{s})\geq n]$ is large for every $n\leq e^{n[H_Q(X)-R_{\mathrm{w}}-R_{\mathrm{s}}-\epsilon]}$ (for an arbitrarily small $\epsilon>0$ and large n), and so, the minimum between 1 and $e^{nR_{\mathrm{s}}}\Pr\left[N(Q_X,\boldsymbol{w},\boldsymbol{s})\geq n\right]$ is certainly 1. Hence, these terms, of the summation over n, contribute altogether a quantity of the exponential order of $e^{n[H_Q(X)-R_{\mathrm{w}}-R_{\mathrm{s}}]}$. For larger n, $\Pr[N(Q_X,\boldsymbol{w},\boldsymbol{s})\geq n]$ decays super–exponentially, and so, these terms contribute a negligible amount. Consequently, considering the factor of $e^{nR_{\mathrm{w}}}$ that stems from the summation over $\boldsymbol{w}$, one term that contributes to the expression of the last line above is $\sum_{Q_X\in\mathcal{G}}P_X(\boldsymbol{x}_Q)e^{n[H_Q(X)-R_{\mathrm{s}}]}$, which is of the exponential order of $\exp\{-n\min_{Q_X\in\mathcal{G}}[D(Q_X\|P_X)+R_{\mathrm{s}}]\}$. The other term comes from the types that belong to $\mathcal{G}^c$. For $Q_X\in\mathcal{G}^c$, there are sub–exponentially few terms that contribute $\min\{1,e^{nR_{\mathrm{s}}}\cdot e^{n[H_Q(X)-R_{\mathrm{s}}-R_{\mathrm{w}}]}\}=e^{-n[R_{\mathrm{w}}-H_Q(X)]_+}$, and so, the overall contribution is $\max_{Q_X\in\mathcal{G}^c}e^{nR_{\mathrm{w}}}e^{-n[H_Q(X)+D(Q_X\|P_X)]}e^{-n[R_{\mathrm{w}}-H_Q(X)]_+}$, which is $\exp\{-n\min_{Q_X\in\mathcal{G}^c}[D(Q_X\|P_X)+[H_Q(X)-R_{\mathrm{w}}]_+]\}$. Thus, the overall performance is

$$
\bar{P}_{\mathrm{FA}}\stackrel{\cdot}{\leq}\exp\left(-n\min_{Q_X}[D(Q_X\|P_X)+\min\{R_{\mathrm{s}},[H_Q(X)-R_{\mathrm{w}}]_+\}]\right), \qquad (56)
$$

completing the proof of Theorem 3.

VI. Information Leakage for the Typical Code

In this last section, which is very brief, we provide an outline for the evaluation of the third figure of merit of our model of an authentication system, namely, the secrecy, or the information leakage, $I(\boldsymbol{W};\boldsymbol{S})$, associated with the typical code, $\mathcal{E}$, in the ensemble.

We envision the typical code as a code with the following properties:

1. For any given type class $\mathcal{T}(Q_X)$ whose size is larger than $e^{n(R_{\rm s}+R_{\rm w})}$, the number of members of $\mathcal{T}(Q_X)$ mapped each one of the $e^{n(R_{\rm s}+R_{\rm w})}$ pairs $(\boldsymbol{s},\boldsymbol{w})$ is exactly the same (uniform distribution of $(\boldsymbol{S},\boldsymbol{W})$ within the type), so that $H(\boldsymbol{S},\boldsymbol{W}|\boldsymbol{X}\in\mathcal{T}(Q_X))=n(R_{\rm s}+R_{\rm w})$.

2. For any given type class $\mathcal{T}(Q_X)$ whose size is smaller than $e^{n(R_{\rm s}+R_{\rm w})}$, each member of $\mathcal{T}(Q_X)$ is mapped to a different pair $(\boldsymbol{s},\boldsymbol{w})$, so that $H(\boldsymbol{S},\boldsymbol{W}|\boldsymbol{X}\in\mathcal{T}(Q_X))=\log|\mathcal{T}(Q_X)|$.

The leakage will then be upper bounded as follows:

$$\begin{aligned}
I(\boldsymbol{S};\boldsymbol{W}) &= H(\boldsymbol{S})+H(\boldsymbol{W})-H(\boldsymbol{S},\boldsymbol{W})\\
&\le nR_{\rm s}+nR_{\rm w}-H(\boldsymbol{S},\boldsymbol{W}|\hat{P}_{\boldsymbol{X}})\\
&= n(R_{\rm s}+R_{\rm w})-\boldsymbol{E}\min\Big\{n(R_{\rm s}+R_{\rm w}),\log|\mathcal{T}(\hat{P}_{\boldsymbol{X}})|\Big\}\\
&= \boldsymbol{E}\left\{\Big[n(R_{\rm s}+R_{\rm w})-\log|\mathcal{T}(\hat{P}_{\boldsymbol{X}})|\Big]_+\right\}\\
&\approx n\boldsymbol{E}\Big\{[R_{\rm s}+R_{\rm w}-\hat{H}_{\boldsymbol{X}}(X)]_+\Big\}. \qquad (57)
\end{aligned}$$

Now, assuming that $H(X)>R_{\rm s}+R_{\rm w}$, the probability of falling in a type class $\mathcal{T}(\hat{P}_{\boldsymbol{x}})$ with $R_{\rm s}+R_{\rm w}-\hat{H}_{\boldsymbol{x}}(X)>0$ is of the exponential order of $\exp\{-nE_{\rm sec}(R_{\rm s}+R_{\rm w})\}$, where

$$E_{\rm sec}(R)\stackrel{\Delta}{=}\min\{D(Q_X\|P_X):\ H_Q(X)\le R\}, \qquad (58)$$

and therefore,

$$\begin{aligned}
I(\boldsymbol{S};\boldsymbol{W}) &\stackrel{\cdot}{\le} n\sum_{\boldsymbol{x}}P_X(\boldsymbol{x})[R_{\rm s}+R_{\rm w}-\hat{H}_{\boldsymbol{x}}(X)]\cdot\mathcal{I}\{R_{\rm s}+R_{\rm w}-\hat{H}_{\boldsymbol{x}}(X)>0\}\\
&\le n(R_{\rm s}+R_{\rm w})\cdot\Pr\{R_{\rm s}+R_{\rm w}-\hat{H}_{\boldsymbol{X}}(X)>0\}\\
&\stackrel{\cdot}{=} \exp\{-nE_{\rm sec}(R_{\rm s}+R_{\rm w})\}, \qquad (59)
\end{aligned}$$

which means that as long as $H(X) > R_{\rm s} + R_{\rm w}$, strong security is guaranteed in the sense that $I(\boldsymbol{S};\boldsymbol{W})$ tends to zero even without normalization by n, as it decays exponentially fast. The secrecy exponent depends on $R_{\rm s}$ and $R_{\rm w}$ only via their sum, $R_{\rm s} + R_{\rm w}$.

Appendix

Proof of eq. (36). The proof is similar to the proof of a similar argument in the context of channel coding [10, Appendix B]. First, observe that

$$Z_{\boldsymbol{x}}(\boldsymbol{y}) = \sum_{\boldsymbol{x}' \neq \boldsymbol{x}} \exp\{na(\hat{P}_{\boldsymbol{x}'\boldsymbol{y}})\} \cdot \mathcal{I}\{f(\boldsymbol{x}') = f(\boldsymbol{x})\} = \sum_{Q_{X|Y}} e^{na(Q_{XY})} N(\mathcal{T}(Q_{X|Y}|\boldsymbol{y}), f(\boldsymbol{x})). \quad \text{(A.1)}$$

Thus, considering the randomness of $\{f(\boldsymbol{x})\}$,

$$\begin{aligned}
& \Pr\left\{Z_{\boldsymbol{x}}(\boldsymbol{y}) \leq \exp\{n\alpha(R+\epsilon, \hat{P}_{\boldsymbol{y}})\}\right\} \\
= \; & \Pr\left\{\sum_{Q_{X|Y}} N(\mathcal{T}(Q_{X|Y}|\boldsymbol{y}), f(\boldsymbol{x}))e^{na(Q_{XY})} \leq \exp\{n\alpha(R+\epsilon, \hat{P}_{\boldsymbol{y}})\}\right\} \\
\leq \; & \Pr\left\{\max_{Q_{X|Y}} N(\mathcal{T}(Q_{X|Y}|\boldsymbol{y}), f(\boldsymbol{x}))e^{na(Q_{XY})} \leq \exp\{n\alpha(R+\epsilon, \hat{P}_{\boldsymbol{y}})\}\right\} \\
= \; & \Pr\bigcap_{Q_{X|Y}} \left\{N(\mathcal{T}(Q_{X|Y}|\boldsymbol{y}), f(\boldsymbol{x}))e^{na(Q_{XY})} \leq \exp\{n\alpha(R+\epsilon, \hat{P}_{\boldsymbol{y}})\}\right\} \\
= \; & \Pr\bigcap_{Q_{X|Y}} \left\{N(\mathcal{T}(Q_{X|Y}|\boldsymbol{y}), f(\boldsymbol{x})) \leq \exp\{n[\alpha(R+\epsilon, \hat{P}_{\boldsymbol{y}}) - a(Q_{XY})]\}\right\}. \quad \text{(A.2)}
\end{aligned}$$

Now, $N(\mathcal{T}(Q_{X|Y}|\boldsymbol{y}), f(\boldsymbol{x}))$ is a binomial random variable with $|\mathcal{T}(Q_{X|Y}|\boldsymbol{y}) \doteq e^{nH_Q(X|Y)}$ trials and success rate of $e^{-nR_{\rm w}}$. We now argue that by the very definition of $\alpha(R+\epsilon, \hat{P}_{\boldsymbol{y}})$, there must exist some $Q^*_{X|Y}$ such that for $Q^*_{XY} = \hat{P}_{\boldsymbol{y}} \times Q^*_{X|Y}$, $H_{Q^*}(X|Y) \geq R + \epsilon$ and $H_{Q^*}(X|Y) - R - \epsilon \geq \alpha(R+\epsilon, \hat{P}_{\boldsymbol{y}}) - a(\hat{P}_{\boldsymbol{y}} \times Q^*_{X|Y})$. Let then $Q^*_{X|Y}$ be such a conditional distribution. Then,

$$\begin{aligned}
& \Pr\bigcap_{Q} \left\{N(\mathcal{T}(Q_{X|Y}|\boldsymbol{y}), f(\boldsymbol{x})) \leq \exp\{n[\alpha(R+\epsilon, \hat{P}_{\boldsymbol{y}}) - a(\hat{P}_{\boldsymbol{y}} \times Q_{X|Y})]\}\right\} \\
\leq \; & \Pr\left\{N(\mathcal{T}(Q^*_{X|Y}|\boldsymbol{y}), f(\boldsymbol{x})) \leq \exp\{n[\alpha(R+\epsilon, \hat{P}_{\boldsymbol{y}}) - a(\hat{P}_{\boldsymbol{y}} \times Q^*_{X|Y})]\}\right\}. \quad \text{(A.3)}
\end{aligned}$$

Now, we know that $H_{Q^*}(X|Y) \geq R + \epsilon$ and $H_{Q^*}(X|Y) - R - \epsilon \geq \alpha(R+\epsilon, \hat{P}_{\boldsymbol{y}}) - a(\hat{P}_{\boldsymbol{y}} \times Q^*_{X|Y})$. By the Chernoff bound (see, e.g., [9, Chap. 6]), the probability in question is upper bounded by

$$\exp\left\{-e^{nH_{Q^*}(X|Y)} D(e^{-\alpha n} \| e^{-\beta n})\right\}, \quad \text{(A.4)}$$

where $\alpha = H_{Q^*}(X|Y) + a(\hat{P}_{\boldsymbol{y}} \times Q^*_{XY}) - \alpha(R+\epsilon, \hat{P}_{\boldsymbol{y}})$ and $\beta = R$. Noting that $\alpha - \beta \geq \epsilon$, we can easily lower bound the binary divergence as follows (see [9, Section 6.3]):

$$\begin{aligned} D(e^{-\alpha n} \| e^{-\beta n}) &\geq e^{-\beta n}\{1 - e^{-(\alpha-\beta)n}[1 + n(\alpha-\beta)]\} \\ &\geq e^{-nR}[1 - e^{-n\epsilon}(1+n\epsilon)], \end{aligned} \tag{A.5}$$

where in the last passage, we have used the decreasing monotonicity of the function $f(t) = (1+t)e^{-t}$ for $t \geq 0$. Thus,

$$\begin{aligned} & \Pr\left\{N(\mathcal{T}(Q^*_{X|Y}|\boldsymbol{y}), f(\boldsymbol{x})) \leq \exp\{n[\alpha(R, \hat{P}_{\boldsymbol{y}}) - a(\hat{P}_{\boldsymbol{y}} \times Q^*_{X|Y}) - \epsilon]\}\right\} \\ \leq\ & \exp\left\{-e^{nH_{Q^*}(X|Y)} \cdot e^{-nR}[1 - e^{-n\epsilon}(1+n\epsilon)]\right\} \\ \leq\ & \exp\left\{-e^{n\epsilon}[1 - e^{-n\epsilon}(1+n\epsilon)]\right\} \\ =\ & \exp\left\{-e^{n\epsilon} + n\epsilon + 1\right\}. \end{aligned} \tag{A.6}$$

Finally, the factor of $|\mathcal{X} \times \mathcal{Y}|^n$ in eq. (36) comes from the union bound, taking into account all $|\mathcal{X} \times \mathcal{Y}|^n$ possible pairs $\{(\boldsymbol{x}, \boldsymbol{y})\}$. This completes the proof of eq. (36).

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